

14.21

(1) $R=0.2\Omega$ $N=100$



$$\frac{di}{dt} = \frac{20A}{0.01s} = 2 \times 10^3 A \cdot s^{-1}$$

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

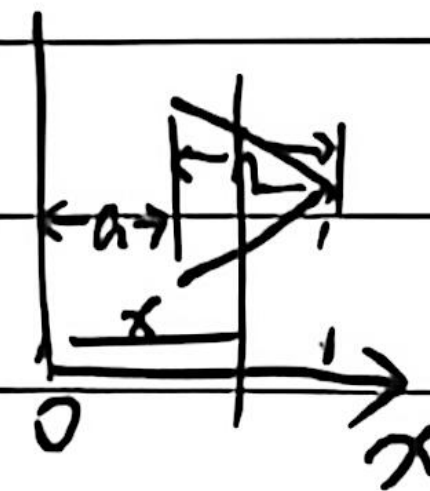
$$B = \frac{\mu_0 N i}{2R}$$

$$\therefore \mathcal{E} = N \frac{d(\mu_0 N i S)}{dt} = \frac{\mu_0 N^2 S}{2R} \frac{di}{dt} = 3.14 \times 10^{-4} V$$

(2) $\mathcal{E} = M \frac{di}{dt} \Rightarrow M = \frac{\mathcal{E}}{\frac{di}{dt}} = \frac{3.14 \times 10^{-4} V}{2 \times 10^3 A \cdot s^{-1}} = 1.57 \times 10^{-7} H$

14.23

$$M = \frac{\Phi}{I} \quad \Phi = \int B(x) \cdot dS$$

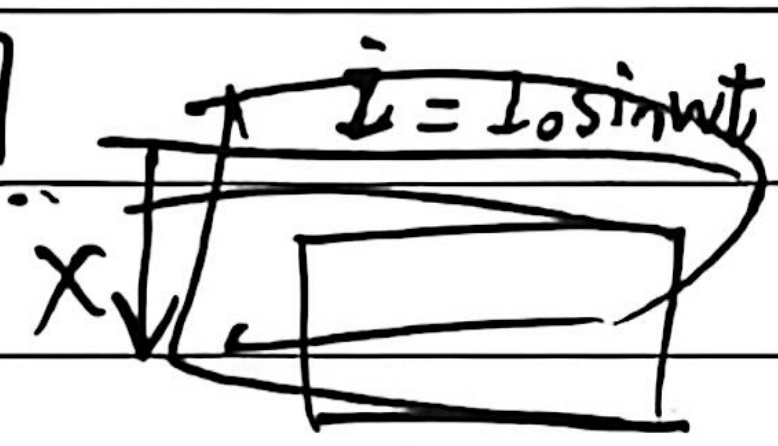


$$B(x) = \frac{\mu_0 I}{2\pi x} \quad dS = dx \cdot \frac{2}{\sqrt{3}} (a+h-x)$$

$$\therefore \Phi = \int_a^{a+h} \frac{\mu_0 I}{2\pi} \frac{2}{\sqrt{3}} \frac{a+h-x}{x} dx = \frac{\mu_0 I}{\sqrt{3}\pi} \left[(a+h) \ln \frac{a+h}{a} - h \right]$$

$$\therefore M = \frac{\Phi}{I} = \frac{\mu_0}{\sqrt{3}\pi} \left[(a+h) \ln \frac{a+h}{a} - h \right]$$

14.25

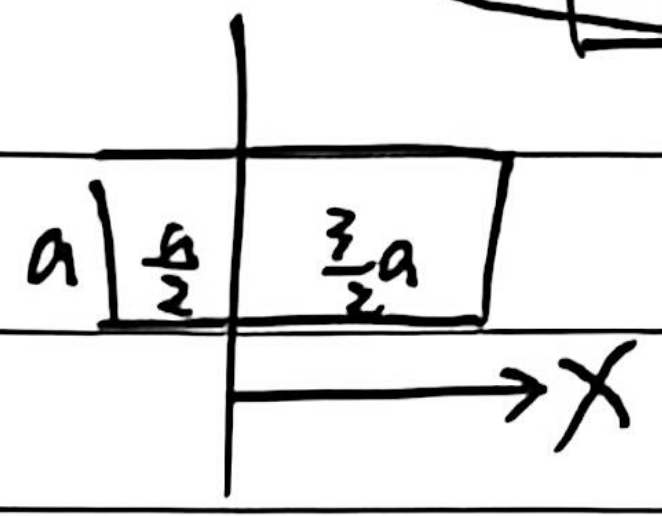


(1) $M = \frac{\Phi}{I_0 \sin wt}$

$$B(x) = \frac{\mu_0 I}{2\pi x}$$

$$\Phi = \frac{\mu_0 I a}{2\pi} \ln 3$$

$$\therefore M = \frac{\mu_0 a}{2\pi} \ln 3$$



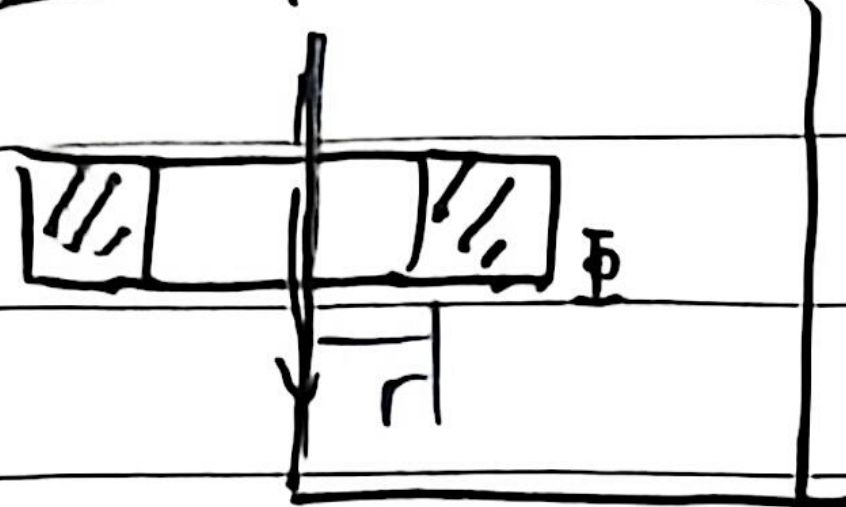
$$\Phi = \int_0^{\frac{3}{2}a} \frac{\mu_0 I}{2\pi x} a dx = \frac{\mu_0 I a}{2\pi} \ln 3$$

(2) $\mathcal{E} = -M \frac{dI}{dt} = \frac{\mu_0 a}{2\pi} \ln 3 \cdot \omega I_0 \cos wt$ 顺时方向

互感电动势符号

KCL, KVL 不符号

14-26



$$B = \frac{\mu_0 N I}{2\pi r}$$

$$\Phi = B \cdot b(R_2 - R_1) = \mu_0 N I b (R_2 - R_1)$$

$$M = \frac{N\Phi}{I} = \mu_0 N^2 b (R_2 - R_1) = 0.012 \text{ (SI)}$$

$$\epsilon = M \frac{dI}{dt} = -M \cdot \omega I_0 \sin \omega t = 13.33 \text{ V}$$

大小 13.33V 方向 顺时针

更好的做法 $M_{12} = M_{21}$

$$\text{直导线通 } I, B(r) = \frac{\mu_0 I}{2\pi r} \quad \Phi = \int_{R_1}^{R_2} \frac{\mu_0 I}{2\pi r} b dr = \frac{\mu_0 I b}{2\pi} \ln \frac{R_2}{R_1}$$

$$M = \frac{N\Phi}{I} = \frac{\mu_0 N b}{2\pi} \ln \frac{R_2}{R_1}$$

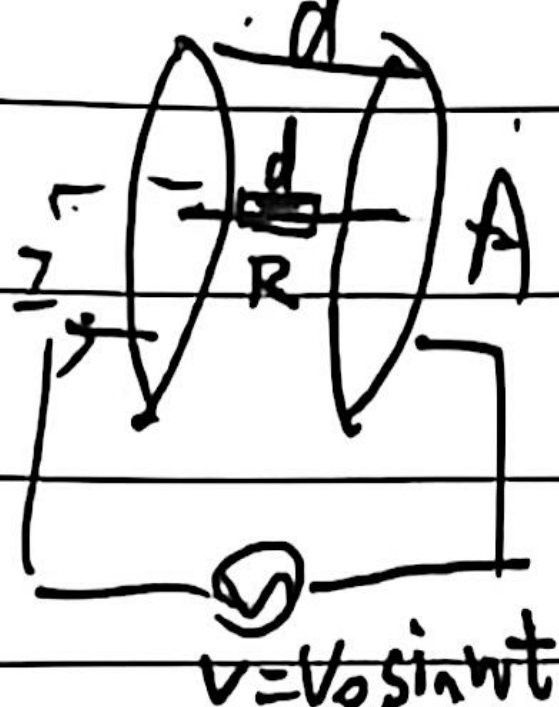
$$\epsilon_m = -M \frac{dI}{dt} = \frac{\mu_0 N b I_0 \omega}{2\pi} \sin \omega t \ln \frac{R_2}{R_1} = 1.47 \times 10^{-2} \text{ V}$$



CS 扫描全能王

3亿人都在用的扫描App

15.3



$$(1) i = \frac{V}{R} = \frac{V_0 \sin \omega t}{R}$$

$$(2) I_d = \frac{d\Phi_D}{dt} \quad E = \frac{U}{d}, \quad D = \epsilon_0 E$$

$$\Phi_D = AD = \frac{A \epsilon_0 V_0 \sin \omega t}{R \cdot d}$$

$$\therefore I_d = \frac{d\Phi_D}{dt} = \frac{A \epsilon_0 V_0}{Rd} \omega \cos \omega t$$

$$(3) I_{\text{总}} = i + I_d = \frac{V_0 \sin \omega t}{R} + \frac{A \epsilon_0 V_0}{Rd} \omega \cos \omega t$$

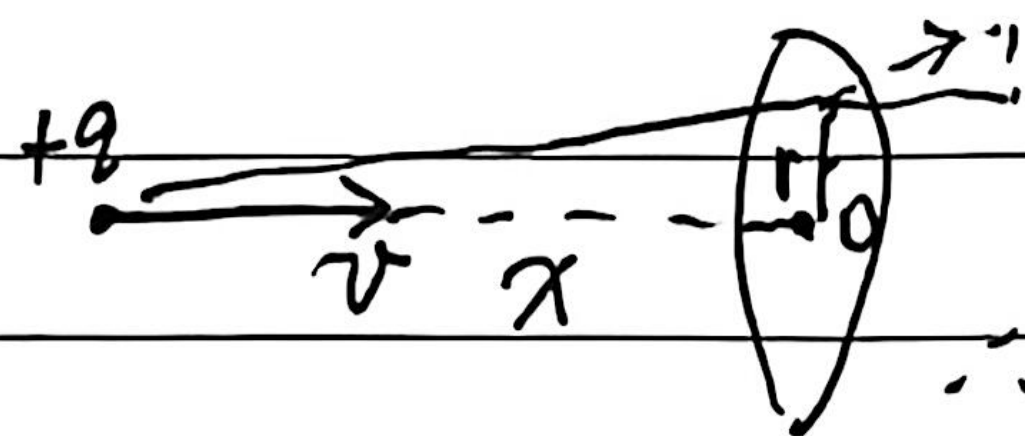
注意没有包含全部位移电流

$$(4) B \cdot 2\pi r = \mu_0 I_{\text{总}}$$

$$B = \frac{\mu_0}{2\pi r} \left[\frac{V_0 \sin \omega t}{R} + \frac{A \epsilon_0 V_0}{Rd} \omega \cos \omega t \right]$$

15.5

$$I_d = \frac{d\Phi_D}{dt} \quad \cancel{I_d = \frac{d\Phi_D}{dt}}$$



$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{x^2 + r^2} \cdot \frac{x}{\sqrt{x^2 + r^2}}$$

$$\therefore \Phi_E = \int_0^a E dr = \frac{qx}{4\pi\epsilon_0} \int_0^a \frac{dr}{(x^2 + r^2)^{3/2}} = \frac{q}{4\pi\epsilon_0 x} \left(\frac{a}{\sqrt{x^2 + a^2}} - \frac{x}{\sqrt{x^2 + a^2}} \right)$$

$$\therefore I_d = \frac{d\Phi_D}{dt} = \epsilon_0 \frac{d\Phi_E}{dt}$$

$$B = \frac{\mu_0}{4\pi} \frac{qv}{r^3} a$$

$$\oint \mu_0 B \cdot dl = \mu_0 B \cdot 2\pi a = \frac{qv a^2}{2(x^2 + a^2)^{3/2}}$$



15.8

$$(1) W = \frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 = \frac{1}{2} \epsilon_0 E_{max}^2 = \frac{1}{2} \mu_0 H_{max}^2$$

$$\Rightarrow \begin{cases} E_{max} = \\ H_{max} = \end{cases}$$

(2) ~~W~~

15.8

$$(1) \bar{S} = \frac{1}{2} E_0 H_0 \quad \sqrt{\epsilon_0} E_0 = \sqrt{\mu_0} H_0$$

$$\therefore \bar{S} = \frac{1}{2} E_0^2 \cdot \frac{\sqrt{\epsilon_0}}{\sqrt{\mu_0}} = \frac{1}{2} H_0^2 \cdot \frac{\sqrt{\mu_0}}{\sqrt{\epsilon_0}}$$

$$\Rightarrow \begin{cases} E_0 = 1.03 \times 10^3 \text{ V/m} \\ B_0 = 3.42 \times 10^{-4} \text{ T} \end{cases}$$

$$B_0 = 3.42 \times 10^{-4} \text{ T}$$

$$(2) P = \bar{S} \cdot 4\pi r^2 = 3.96 \times 10^{26} \text{ J/s}$$

$$H = \frac{1}{\mu_0} B$$

15.11 $\lambda = 3 \quad A = 300 \text{ V/m}$

$$(1) f = \frac{c}{\lambda} = 1 \times 10^8 \text{ s}^{-1} = \frac{1}{\sqrt{\mu_0}} B$$

$$(2) \vec{z} \uparrow \vec{B} \uparrow, \quad \sqrt{\epsilon_0} A_0 = \sqrt{\mu_0} A_H \quad A_B = \sqrt{\frac{\epsilon_0 \mu_0}{\epsilon_0}} A = \frac{A}{c}$$

$$(3) \omega = 2\pi f = 6.28 \times 10^8 \text{ rad/s}$$

$$= 1 \times 10^4 \text{ T}$$

$$(4) \bar{S} = \frac{1}{2} E_0 H_0 = \frac{1}{2} E_0^2 \sqrt{\frac{\epsilon_0}{\mu_0}} = 119 \text{ W/m}^2$$

15.14 $w(t) = \alpha t \quad \vec{i} = q \cdot \vec{v} = \sigma \cdot 2\pi R \cdot L \cdot \frac{w}{2\pi}$ RLS W



$$(1) B = \mu_0 \vec{i} = \mu_0 \sigma \cdot 2\pi R \cdot \alpha t$$

$$(2) E \cdot 2\pi R = \frac{d\Phi}{dt} = \frac{d(B \cdot 2\pi R^2)}{dt} = \mu_0 \sigma R \cdot d \cdot 2\pi R^2$$

$$E = \frac{\mu_0 \sigma R^2 d}{2}$$

$$(3) S = E \cdot H = \frac{E B}{\mu_0} = \frac{\mu_0}{2} R^3 \sigma^2 \alpha^2 t^2$$

$$(4) P = \int S \cdot dA = S \cdot 2\pi R L = \frac{d}{dt} \left(\frac{\pi R^2 L}{2 \mu_0} B^2 \right)$$